OFFSIDE LABS

Dynamic AMM

Release 0.5.4

Smart Contract Security Assessment

May 2026

Prepared for:

Meteora

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Contents

1 About Offside Labs

Offside Labs is a leading security research team, composed of top talented hackers from both academia and industry.

We possess a wide range of expertise in modern software systems, including, but not limited to, *browsers*, *operating systems*, *IoT devices*, and *hypervisors*. We are also at the forefront of innovative areas like *cryptocurrencies* and *blockchain technologies*. Among our notable accomplishments are remote jailbreaks of devices such as the **iPhone** and **PlayStation 4**, and addressing critical vulnerabilities in the **Tron Network**.

Our team actively engages with and contributes to the security community. Having won and also co-organized *DEFCON CTF*, the most famous CTF competition in the Web2 era, we also triumphed in the **Paradigm CTF 2023** within the Web3 space. In addition, our efforts in responsibly disclosing numerous vulnerabilities to leading tech companies, such as *Apple*, *Google*, and *Microsoft*, have protected digital assets valued at over **$300 million**.

In the transition towards Web3, Offside Labs has achieved remarkable success. We have earned over **$9 million** in bug bounties, and **three** of our innovative techniques were recognized among the **top 10 blockchain hacking techniques of 2022** by the Web3 security community.

https://offside.io/

https://github.com/offsidelabs

https://twitter.com/offside_labs

2 Executive Summary

Introduction

Offside Labs completed a security audit of *Dynamic AMM* smart contracts, starting on *May 21th, 2026* and concluding on *May 22th, 2026*.

Project Overview

Dynamic AMM Pools are designed to provide sustainable liquidity for decentralized finance by leveraging a capital allocation layer. Rather than relying on unsustainable liquidity mining incentives, Dynamic AMM deposits assets directly into this layer, where they are dynamically allocated to external lending protocols. This allows LPs to earn yield and rewards on idle assets. By reducing dependence on liquidity mining, Dynamic AMM mitigates issues like token inflation and short-term liquidity loss, instead fostering long-term liquidity provision. This approach emphasizes value creation and aims to attract committed LPs to the ecosystem.

Updates Summary

Dynamic AMM Release 0.5.4 introduces protocol fee claiming through a dedicated protocol fee authority, allowing one pool token to be claimed per call while preserving partner fee balances. It deprecates the previous operator controlled withdrawal and zap paths in favor of the protocol fee wrapper program, and adds claim events that record the pool, receiver, token mint, and amount.

Audit Scope

The assessment scope contains mainly the smart contracts of the amm program for the *Dynamic AMM* project.

The audit is based on the following specific branches and commit hashes of the codebase repositories:

- Meteora Dynamic AMM
  - Codebase: `https://github.com/MeteoraAg/meteora-dynamic-amm`
  - Release 0.5.4: PR-213
  - Commit Hash: 571cb6f3d58677deb666a0eca5004bf0e0164148

We listed the files we have audited below:

- Meteora Dynamic AMM Release 0.5.4
  - programs/amm/src/const_pda.rs
  - programs/amm/src/constants.rs
  - programs/amm/src/depeg/marinade.rs
  - programs/amm/src/depeg/solido.rs
  - programs/amm/src/error.rs
  - programs/amm/src/event.rs

- programs/amm/src/instructions/claim_protocol_fee2.rs
- programs/amm/src/instructions/initialize_pool/alpha_vault_utils.rs
- programs/amm/src/lib.rs

Findings

The security audit identified no critical, high, medium, low, or informational issues in the specified PRs. The implementation adheres to security best practices for the assessed components.

3 Detailed Summary

Scope Confirmation

The audit covered:

1. Functionality: adds a new protocol fee claim flow where the protocol fee authority can claim one pool token per instruction through the protocol fee wrapper program, while excluding partner pending fees from the claimable amount.
2. PRs:
   - PR-213
     - Purpose: move protocol fee collection from operator controlled AMM instructions to a dedicated protocol fee authority and wrapper flow, with claims limited to one token per call and partner fee balances protected.

Conclusion

After thorough review, the audited components:

- Follow secure design patterns for the intended functionality covered by the audit;
- Introduce no new security risks within the given threat model.

4 Disclaimer

This audit report is provided for informational purposes only and is not intended to be used as investment advice. While we strive to thoroughly review and analyze the smart contracts in question, we must clarify that our services do not encompass an exhaustive security examination. Our audit aims to identify potential security vulnerabilities to the best of our ability, but it does not serve as a guarantee that the smart contracts are completely free from security risks.

We expressly disclaim any liability for any losses or damages arising from the use of this report or from any security breaches that may occur in the future. We also recommend that our clients engage in multiple independent audits and establish a public bug bounty program as additional measures to bolster the security of their smart contracts.

It is important to note that the scope of our audit is limited to the areas outlined within our engagement and does not include every possible risk or vulnerability. Continuous security practices, including regular audits and monitoring, are essential for maintaining the security of smart contracts over time.

Please note: we are not liable for any security issues stemming from developer errors or misconfigurations at the time of contract deployment; we do not assume responsibility for any centralized governance risks within the project; we are not accountable for any impact on the project's security or availability due to significant damage to the underlying blockchain infrastructure.

By using this report, the client acknowledges the inherent limitations of the audit process and agrees that our firm shall not be held liable for any incidents that may occur subsequent to our engagement.

This report is considered null and void if the report (or any portion thereof) is altered in any manner.

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